S&T2025

Bridging Physics

Tutorial 5

2026

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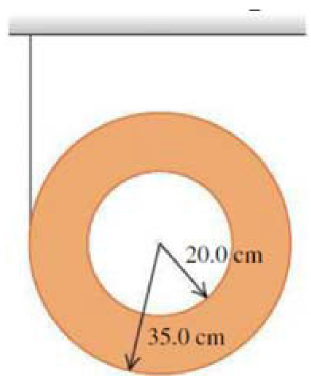
1. A thin, light string is wrapped around the outer rim of a uniform hollow cylinder of mass 4.80 kg having inner and outer radii, as shown in figure.



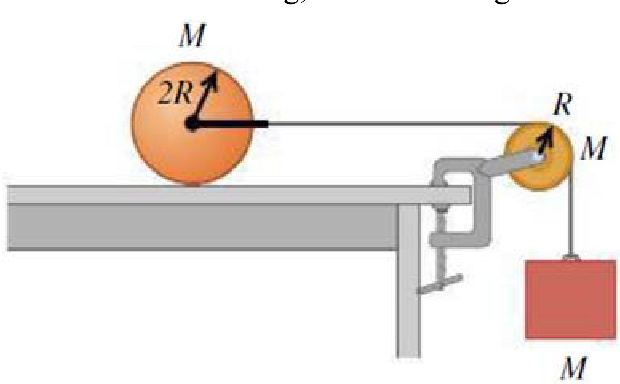
The cylinder is released from rest.

(i) How far must the cylinder fall before its centre of mass is moving at 6.44 m/s? (3.52 m)

(ii) If you just dropped this cylinder without any string, how fast would its centre of mass be moving when it had fallen the distance in part (i)? (8.31 m/s)

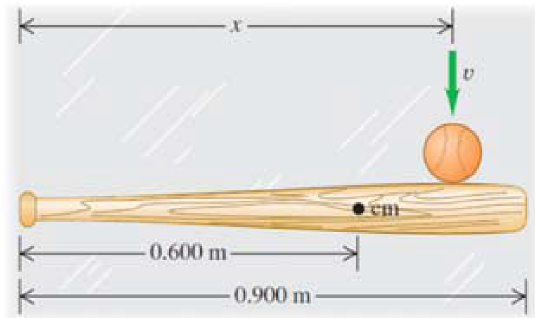
(iii) Why do you get two different answers when the cylinder falls the same distance in both cases?

2. A uniform solid cylinder with mass M and radius $2R$ rests on a horizontal tabletop. A string is attached by a yoke to a frictionless axle through the centre of the cylinder so that the cylinder can rotate about the axle. The string runs over a disk-shaped pulley with mass M and radius R that is mounted on a frictionless axle through its centre. A block of mass M is suspended from the free end of the string, as shown in figure.



The string doesn't slip over the pulley surface, and the cylinder rolls without slipping on the tabletop. Find the magnitude of the acceleration of the block after the system is released from rest. $\left(\frac{g}{3}\right)$

3. A baseball bat rests on a frictionless, horizontal surface. The bat has a length of 0.900 m, a mass of 0.800 kg, and its centre of mass is 0.600 m from the handle end of the bat, as shown in figure.



The moment of inertia of the bat about its centre of mass is 0.0530 kg m^2. The bat is struck by a baseball travelling perpendicular to the bat. The impact applies an impulse at a point a distance x from the handle end of the bat. What must x be so that the handle end of the bat remains at rest as the bat begins to move? (0.7104 m)

Remark. The point on the bat you have located is called the centre of percussion. Hitting a pitched ball at the *centre of percussion* of the bat minimises the "sting" the batter experiences on the hands.

4. A 2.00-mol sample of oxygen gas is confined to a 5.00-L vessel at a pressure of 8.00 atm. Find the average translational kinetic energy of the oxygen molecules under these conditions. $\left(5.05\times10^{-21}\text{ J}\right)$

5. A vertical cylinder with a heavy piston contains air at 300K. The initial pressure is 2×10^5 Pa, and the initial volume is 0.350 m^3. Take the molar mass of air as 28.9 g/mole and assume $C_V = \frac{5}{2}R.$

(i) Find the specific heat of air at constant volume in units of J/kg$\bullet$°C.
$\left(719.20\ \frac{\text{J}}{\text{kg}\cdot\text{K}}\right)$

(ii) Calculate the mass of the air in the cylinder. (0.8111 kg)

(iii) Suppose the piston is held fixed. Find the energy input required to raise the temperature of the air to 700 K. (233.34 kJ)

(iv) What if? Assume again the condition of the initial state and assume the heavy piston is free to move. Find the energy input required to raise the temperature to 700 k. (326.67 kJ)

6. A certain monatomic ideal gas has molar heat capacity at constant volume C_V. A sample of this gas initially occupies a volume V_0 at pressure p_0 and absolute temperature T_0. The gas expands isobarically to a volume $2V_0$ and then expands further *adiabatically* to a final volume $4V_0$. (You can make use of $\gamma = \dfrac{C_V + R}{C_V}$.)

(i) Draw a pV-diagram for this sequence of processes.

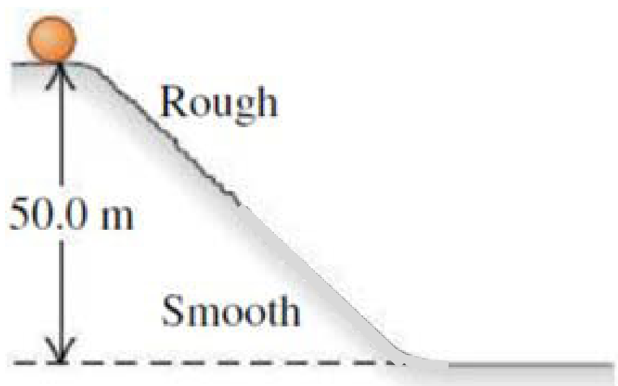
(ii) Find the final temperature of the gas. $\left(\dfrac{T_0}{2^{\gamma\text{-}2}}\right)$

(iii) Compute the total work done by the gas for this sequence of processes. $\left(\dfrac{3}{2}\left(\dfrac{8}{3}-\dfrac{1}{2^{\gamma\text{-}2}}\right)P_0V_0\right)$

(iv) Find the absolute value of the total heat flow $|Q|$ into or out of the gas for this sequence of processes, and state the direction of heat flow. $\left(P_0V_0\left(1+\dfrac{C_V}{R}\right)\right)$

7.

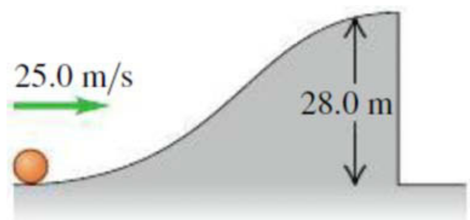
(i) A solid, uniform, spherical boulder starts from rest and rolls down a 50.0 m high hill as shown in the figure.



The top half of the hill is rough enough to cause the boulder to roll without slipping, but the lower half is covered with slippery ice and there is no friction. What is the translational speed of the boulder when it reaches the bottom of the hill? (28.98 m/s)

(ii) A basketball (which can be closely modelled as a hollow spherical shell) rolls down a mountainside into a valley and then up the opposite side, starting from rest at a height H_0 above the bottom. In the figure the rough part of the terrain prevents slipping while the smooth part has no friction. Will the ball rise to the same height of H_0 on the other side? Justify your answer with calculation.

(iii) A solid uniform ball rolls without slipping up a hill, as shown in figure.

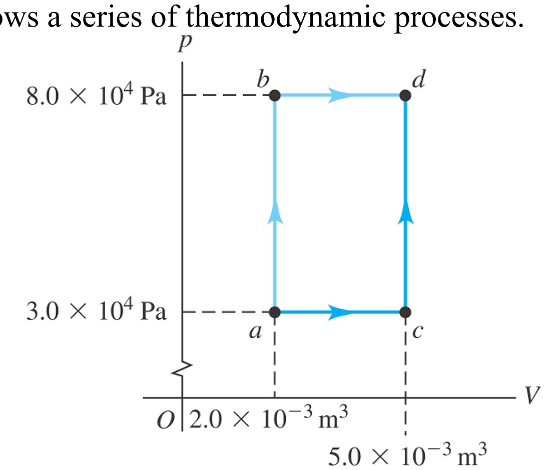


At the top of the hill, it is moving horizontally, and then it goes over the vertical cliff.

(a) How far from the foot of the cliff does the ball land, and how fast is it moving just before it lands? (36.49 m)

(b) Notice that when the balls lands, it has a greater translational speed than when it was at the bottom of the hill. Is this an evidence to violate the law on conservation of energy? Explain.

8. The pV-diagram shows a series of thermodynamic processes.



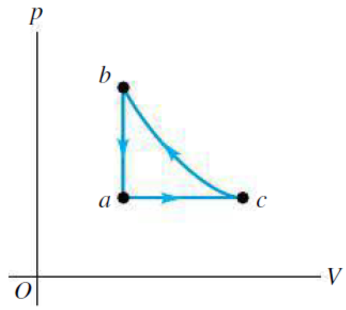
In process $a \rightarrow b$, 150 J of heat is added to the system;
in process $b \rightarrow d$, 600 J of heat is added. Find

(i) the internal energy change in process $a \rightarrow b$; (150 J)

(ii) the internal energy change in process $a \rightarrow b \rightarrow d$; and (510 J)

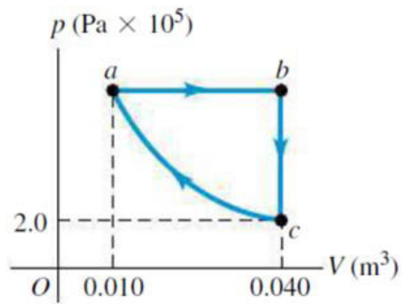
(iii) the total heat added in process $a \rightarrow c \rightarrow d$. (600 J)

9. Three moles of an ideal gas are taken around cycle *abc* as shown in figure.



For this gas, C_p = 29.1 J mol^{-1} K^{-1} . Process *ac* is at constant pressure, process *ba* is at constant volume, and process *cb* is adiabatic. The temperatures of the gas in states *a*, *c* and *b* are T_a = 300 K, T_c = 492 K and T_b = 600 K. Calculate the total work *W* for the cycle. (-1945.8 J)

10. The graph in figure shows a *pV*-diagram for 3.25 mol of ideal helium (He) gas. Part *ca* of this process is *isothermal*.



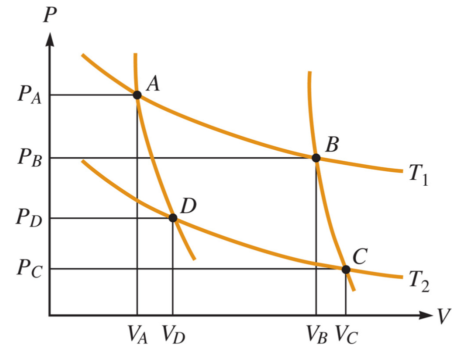
(i) Find the pressure of the He at point *a*. $\left(8\times10^5 \text{ Pa}\right)$

(ii) Find the temperature of the He at points *a*, *b*, and *c*. (296.07 K, 1184.29K)

(iii) How much heat entered or left the He during segments *ab*, *bc*, and *ca*? In each segment, did the heat enter or leave? (60000.37 J, -36000.22 J, -11090.30 J)

(iv) By how much did the internal energy of the He change from *a* to *b*, from *b* to *c*, and from *c* to *a*? Indicate whether this energy increased or decreased.
(36000.22 J, -36000.22 J, 0)

11. In an adiabatic process, the relationship between pressure and volume is PV^{γ} = constant, where γ is the adiabatic exponent that depends on the properties of the gas. Two different adiabatic processes *AD* and *BC* intersect two isothermal processes *AB* at temperature T_1 and *DC* at temperature T_2 as shown in the figure. Is the ratio $\frac{V_A}{V_D}$ greater than, less than, or equal to the ratio $\frac{V_B}{V_C}$?



12. An air rifle shoots a lead pellet by allowing high-pressure air to expand, propelling the pellet down the rifle barrel. Because this process happens very quickly, no appreciable thermal conduction occurs, and the expansion is essentially adiabatic. Suppose that the rifle starts by admitting to the barrel 12.0 cm^3 of compressed air of initial pressure p_i in the barrel, which behaves as an ideal gas with $\gamma = 1.40$. The air expands behind a 1.10-g pellet and pushes on it as a piston with cross-sectional area 0.0300 cm^2, as the pellet moves 50.00 cm along the gun barrel. The pellet emerges with muzzle speed 120 m/s. Calculate the initial pressure required. $\left(5.74\times10^{6}\ \dfrac{\mathrm{N}}{\mathrm{m}^2}\right)$

- End -

Formula List

Logarithm:

$$\log a + \log b = \log(a \times b)$$
$$\log a - \log b = \log\left(\frac{a}{b}\right)$$
$$(\log x) > 0, \text{ if } x > 1.$$
$$(\log 1) = 0.$$
$$(\log x) < 0, \text{ if } 0 < x < 1.$$
$$(\log x) \text{ is undefined if } x < 0.$$

Trigonometry:

$$\sin = \frac{opposite}{hypotenuse}$$
$$\cos = \frac{adjacent}{hypotenuse}$$
$$\tan = \frac{opposite}{adjacent} = \frac{\sin}{\cos}$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$
$$\cos 2\theta = 1 - 2\sin^2\theta$$
$$\cos 2\theta = 2\cos^2\theta - 1$$
$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

Differentiation:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$
$$\frac{d}{dx}(e^x) = e^x$$
$$\frac{d}{dx}\left(e^{f(x)}\right) = e^{f(x)} \times \frac{d}{dx}(f(x))$$
$$\frac{d}{dx}(\sin x) = \cos x$$
$$\frac{d}{dx}(\cos x) = -\sin x$$
$$\frac{d}{dx}(f(x) \times g(x)) = f(x) \times \left(\frac{d}{dx}g(x)\right) + \left(\frac{d}{dx}f(x)\right) \times g(x)$$

Integration:

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$$

Vector:

$$\vec{\mathbf{A}} + \vec{\mathbf{B}} = \vec{\mathbf{B}} + \vec{\mathbf{A}}$$
$$\vec{\mathbf{A}} \bullet \vec{\mathbf{B}} = |\mathbf{A}| \times |\mathbf{B}| \times \cos\theta$$

Linear kinematic equations:

$$v_f = v_i + at$$
$$\Delta x = v_i t + \frac{1}{2}at^2$$
$$\Delta x = \frac{(v_i + v_f)}{2} \times t$$
$$v_f^2 = v_i^2 + 2a\Delta x$$

Simple harmonic motion:

$$x(t) = r_0 \cos(\theta_0 + \omega_0 t)$$
$$v_x(t) = -r_0\omega_0 \sin(\theta_0 + \omega_0 t)$$
$$a_x(t) = -r_0\omega_0^2 \cos(\theta_0 + \omega_0 t) = -\omega_0^2 x(t)$$
$$\omega^2 = \frac{k}{m}$$
$$T = 2\pi\sqrt{\frac{m}{k}}$$
$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$
$$v_{max} = \sqrt{\frac{k}{m}}A$$
$$a_{max} = \frac{k}{m}A$$

Retarding force:

$$R = -bv$$

Projectile motion:

$$y = -\frac{g}{2{v_{0x}}^2}\left(x - x_0\right)^2 + \frac{v_{0y}}{v_{0x}}\left(x - x_0\right) + y_0 \text{, or}$$

$$y = x\tan\theta - \frac{gx^2}{2{v_0}^2}\left(1 + \tan^2\theta\right)$$

Uniform circular motion:

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T}$$

$$v = r\omega$$

$$a_c = r\omega^2 = \frac{v^2}{r}$$

Non-uniform circular motion:

$$a_t = \frac{dv}{dt}$$

$$a = \sqrt{a_c^2 + a_t^2}$$

Relative velocity:

$$\vec{\mathbf{V}}_{P/A} = \vec{\mathbf{V}}_{P/B} + \vec{\mathbf{V}}_{B/A}$$

Newton's second law:

$$\sum \vec{F} = m\vec{a}$$

Newton's third law:

$\vec{F}_{12} = -\vec{F}_{21}$, and the two forces act on the opposite body.

Frictional force:

$f_s \le \mu_s n,$ and impending motion occurs when $f_s = \mu_s n.$

$$f_k = \mu_k n$$

Translational kinetic energy:

$$KE_{\text{translational}} = \frac{1}{2}mv^2$$

Gravitational potential energy:

$$\Delta PE = mg(y_2 - y_1) = mgh$$

Spring

$$F_{\text{spring}} = -kx$$

Elastic $PE_{\text{spring}} = \frac{1}{2}kx^2$

Work-energy theorem:

$$W = \Delta KE = KE_f - KE_i$$

Work done (constant force):

$$W = \vec{F} \bullet \Delta\vec{r} = F\Delta r\cos\theta$$

Work done (varying force):

$$W = \int_{x_i}^{x_f} \vec{F} \bullet d\vec{r}$$

Power:

$$P = \frac{dW}{dt} = F \times v$$

Linear momentum:
(also valid for y and z)

$$p_x = mv_x$$

$$F_x = \frac{dp_x}{dt}$$

$$F_{\text{average}, x} = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} F_x(t)\, dt$$

Conservation of linear momentum:

$$\vec{p}_{1,i} + \vec{p}_{2,i} = \vec{p}_{1,f} + \vec{p}_{2,f}$$

Elastic collision:

$$v_{A1} - v_{B1} = -(v_{A2} - v_{B2})$$

$$v_{A2} = \left(\frac{m_A - m_B}{m_A + m_B}\right)v_{A1} + \left(\frac{2m_B}{m_A + m_B}\right)v_{B1}$$

$$v_{B2} = \left(\frac{m_B - m_A}{m_A + m_B}\right)v_{B1} + \left(\frac{2m_A}{m_A + m_B}\right)v_{A1}$$

Center of mass
(also valid for y and z)

$$x_{cm} = \frac{1}{M}\sum mx, \text{ where } M = \sum m, \text{ or}$$
$$= \frac{1}{M}\int x\, dm$$
$$v_{cm,x} = \frac{1}{M}\sum mv_x$$
$$a_{cm,x} = \frac{1}{M}\sum ma_x$$

Simple Harmonic Motion:

$$a_x = -\omega^2 x$$
$$\omega^2 = \frac{k}{m}$$
$$T = 2\pi\sqrt{\frac{m}{k}}$$
$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$
$$v_{max} = \sqrt{\frac{k}{m}}A$$
$$a_{max} = \frac{k}{m}A$$

Retarding force:
$$R = -bv$$

Rotation

$$\theta = \frac{s}{r}$$
$$\omega = \frac{d\theta}{dt}$$
$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$
$$v_{\text{tangential}} = r\omega$$
$$a_{\text{centripetal}} = r\omega^2 = \frac{v^2}{r}$$
$$a_{\text{tangential}} = r\alpha$$

Rotational kinematic equations:

$$\omega_f = \omega_i + \alpha t$$
$$\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$$
$$\Delta\theta = \frac{(\omega_i + \omega_f)}{2} \times t$$
$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

Moment of inertia:

$$I = \sum mr^2$$
$$I = \int r^2 dm$$

Parallel axis theorem

$$I = I_{cm} + Md^2$$

Rotational kinetic energy:

$$KE_{\text{rotational}} = \frac{1}{2}I\omega^2$$

Moment of inertia:

$$I = \sum m_i r_i^2$$
$$I = \int r^2 dm$$

Moment of inertia of hoop or thin cylindrical shell:

$$I_{CM} = MR^2$$

Moment of inertia of hollow cylinder:

$$I_{CM} = \frac{1}{2}M\left(R_1^2 + R_2^2\right)$$

Moment of inertia of solid cylinder:

$$I_{CM} = \frac{1}{2}MR^2$$

Moment of inertia of rectangular plate:

$$I_{CM} = \frac{1}{12}M\left(a^2 + b^2\right)$$

Moment of inertia of thin rod about its CM:

$$I_{CM} = \frac{1}{12} ML^2$$

Moment of inertia of thin rod about its end:

$$I = \frac{ML^2}{3}$$

Moment of inertia of solid sphere:

$$I_{CM} = \frac{2}{5} MR^2$$

Moment of inertia of spherical shell:

$$I_{CM} = \frac{2}{3} MR^2$$

Torque and Moment of force:

$$\tau = F \times r \times \sin\theta$$

$$\sum \tau = I\alpha$$

Work due to rotation:

$$\Delta W = \tau \times \Delta\theta$$

$$W = \int_{\theta_1}^{\theta_2} \tau \, d\theta$$

Power due to rotation:

$$P = \tau \times \omega$$

Pure rolling:

$$v_{cm} = r\omega$$

$$a_{cm} = r\alpha$$

$$KE_{\text{total}} = \frac{1}{2} M v_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \omega^2$$

Conservation of Angular momentum:

$$\underbrace{\sum I\omega}_{\text{Before collision}} = \underbrace{\sum I\omega}_{\text{After collision}}$$

Avogadro's number:

$$N_A \approx 6.022 \times 10^{23} \text{ particles/mole}$$

Ideal gas constant:

$$R \approx 8.314 \text{ J mole}^{-1} \text{ K}^{-1}$$

Boltzmann constant:

$$k = \frac{R}{N_A} \approx 1.381 \times 10^{-23} \text{ J/K}$$

Ideal gas equation:

$$pV = nRT$$

Speed of a molecule

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

Mole

$$n = \frac{N}{N_A}$$

$$n = \frac{m}{M}$$

Absolute temperature

$$T = T_{\text{Celsius}} + 273.15$$

Specific heat *c* and molar heat *C*

$$Q = Mc(T_2 - T_1)$$
$$= nC(T_2 - T_1)$$

Latent heat *L*

$$Q = L \times M$$

First law of thermodynamics

$$\Delta U = Q - W$$

Internal energy for *N* molecules

monatomic: $U = N \times \frac{3}{2} kT$

diatomic: $U = N \times \frac{5}{2} kT$

Relationship between C_p and C_v for an ideal Gas

$$C_p - C_v = R$$

Isochoric process

monatomic: $C_V = \frac{3}{2}R$

diatomic: $C_V = \frac{5}{2}R$

C_p is undefined for isochoric process.

Isobaric process

monatomic: $C_p = \frac{5}{2}R$

diatomic: $C_p = \frac{7}{2}R$

C_V is undefined for isobaric process.

Isothermal process

$$W = nRT\ln\left(\frac{V_f}{V_i}\right)$$

Adiabatic process

$$\gamma = \frac{C_p}{C_V}$$

monatomic: $\gamma = \frac{5}{3}$

diatomic: $\gamma = \frac{7}{5}$

$$P_1V_1^{\gamma} = P_2V_2^{\gamma}$$

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$p_1^{1-\gamma}T_1^{\gamma} = p_2^{1-\gamma}T_2^{\gamma}$$

Thermal efficiency of a heat engine

$$e = 1 - \frac{|Q_C|}{|Q_H|}$$

Coefficient of performance of a heat pump for cooling

$$\text{COP} = \frac{|Q_C|}{|Q_H| - |Q_C|}$$

Entropy

$$\Delta S = \int_{\text{initial state}}^{\text{final state}} \frac{dQ}{T} \geq 0$$

Coulomb's Law:

$$F = \frac{kq_1q_2}{r^2}$$

where

$$k = 8.9875\times10^9 \text{ Nm}^2/\text{c}^2$$

Electric field:

$$E = \frac{F}{q_0}$$

Electric field for point charge:

$$E = \frac{kq}{r^2}$$

Electric field at a distance along central axis of charged ring:

$$E = \frac{kQx}{(x^2 + a^2)^{3/2}}$$

Electric potential:

$$V = \frac{U}{q_0}$$

Electric potential in uniform field:

$$\Delta V = -Ed$$

Electric potential for point charge:

$$V = \frac{kq}{r}$$

Electric potential at a distance along central axis of charged ring:

$$V = \frac{kQ}{\sqrt{x^2 + a^2}}$$

Electric potential energy (for multiple point charges):

$$U = k\left(\frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \frac{q_2q_3}{r_{23}}\right)$$

Electric current:

$$I = \frac{dQ}{dt}$$

$$I_{av} = nqv_dA$$

Ohm's Law:

$$I = \frac{\Delta V}{R}$$

Resistance of ohmic conductor:

$$R = \rho\frac{L}{A}$$

Resistivity:

$$\rho = \rho_0[1 + \alpha(T - T_0)]$$

Electric power:

$$P = I\Delta V = I^2R = \frac{\Delta V^2}{R}$$

Electric energy:

$$U = P\Delta t$$

Terminal voltage:

$$\Delta V = \varepsilon - Ir$$

Resistors in series:

$$R_{eq} = R_1 + R_2 + R_3 + \cdots$$

Resistors in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$$

Kirchhoff's Junction Rule:

$$\Sigma I_{in} = \Sigma I_{out}$$

Kirchhoff's Loop Rule:

$$\Sigma\Delta V = 0$$

Magnetic force on a moving charge:

$$F_B = qvB\sin\theta$$

Magnetic force on a current carrying conductor:

$$F = IlB\sin\theta$$

Radius of moving charged particle in uniform magnetic field:

$$r = \frac{mv}{qB}$$

Angular speed:

$$\omega = \frac{qB}{m}$$

Period:

$$T = \frac{2\pi m}{qB}$$

Velocity selector:

$$v = \frac{E}{B}$$

Magnetic field due to long straight conductor:

$B = \frac{\mu_0 I}{2\pi a}$ where

$$\mu_0 = 4\pi \times 10^{-7}\ \text{Tm/A}$$

Magnetic field at center of circular wire loop:

$$B = \frac{\mu_0 I}{2R}$$

Magnetic field at a distance along central axis of current-carrying ring:

$$B = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}}$$

Magnetic force between parallel conductors:

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

Magnetic field inside toroid:

$$B = \frac{\mu_0 N I}{2\pi r}$$

Magnetic field inside solenoid:

$$B = \mu_0 n I$$

Magnetic flux:

$$\Phi_B = BA\cos\theta$$

Faraday's Law:

$$\varepsilon = -N\frac{d\Phi_B}{dt}$$

Motional emf:

$$I = \frac{Blv}{R}$$

Power input:

$$P = F_{app}v = IlBv$$